CYCOLOY™ Resin C1200HF - Asia

Polycarbonate + ABS

SABIC

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Technical Data

Product Description			
PC+ABS, excellent flow/impact/high heat resistance. Low temperature ductility.			

General			
Material Status	• Commercial: Active		
UL Yellow Card [1]	• E45587-236932 • E207780-228463		
Search for UL Yellow Card	• SABIC • CYCOLOY™ Resin		
Availability	• Asia Pacific		
Uses	• Additive Manufacturing (3D Printing) • Appliances • Automotive Applications • Automotive Exterior Parts • Automotive Interior Parts • Automotive Lighting • Automotive Under the Hood • Construction Applications • Decorative Parts • Electrical Parts • Electrical/Electronic Applications	• Electronic Displays • Fluid Handling • Glazing • Heavy Transportation • Industrial Applications • Lawn and Garden Equipment • Lighting Applications • Material Handling • Medical Devices • Medical/Healthcare Applications • Military/Defense Applications	• Non-specific Food Applications • Optical Applications • Outdoor Applications • Personal Care • Pharmaceuticals • Rail Applications • Recreational Vehicle Applications • Sporting Goods • Water Management
Also Available In	• Europe	• Latin America	• North America

Physical	Nominal Value	Unit	Test Method
Density / Specific Gravity	1.15	g/cm³	ASTM D792
Melt Mass-Flow Rate (MFR) (260°C/5.0 kg)	19	g/10 min	ASTM D1238
Molding Shrinkage - Flow (3.20 mm)	0.50 to 0.70	%	Internal Method
Mechanical	**Nominal Value**	**Unit**	**Test Method**
Tensile Modulus [3]	2270	MPa	ASTM D638
Tensile Strength [4] (Yield)	57.0	MPa	ASTM D638
Tensile Elongation [4]			ASTM D638
Yield	5.0	%	
Break	100	%	
Flexural Modulus [5] (50.0 mm Span)	2340	MPa	ASTM D790
Flexural Strength [5] (Yield, 50.0 mm Span)	88.0	MPa	ASTM D790
Impact	**Nominal Value**	**Unit**	**Test Method**
Notched Izod Impact			ASTM D256
-30°C	480	J/m	
23°C	590	J/m	
Instrumented Dart Impact			ASTM D3763
-30°C, Total Energy	54.0	J	
23°C, Total Energy	54.0	J	
Thermal	**Nominal Value**	**Unit**	**Test Method**
Deflection Temperature Under Load			ASTM D648
0.45 MPa, Unannealed, 3.20 mm	129	°C	
1.8 MPa, Unannealed, 3.20 mm	112	°C	
Vicat Softening Temperature	130	°C	ISO 306/B50
CLTE - Flow (-40 to 40°C)	7.2E-5	cm/cm/°C	ASTM E831
RTI Elec	105	°C	UL 746B
RTI Imp	80.0	°C	UL 746B
RTI Str	105	°C	UL 746B

Form No. TDS-48904-en

Document Created: Thursday, February 10, 2022
Added to Prospector: November 2000
Last Updated: 12/2/2021

Electrical	Nominal Value	Unit	Test Method
Comparative Tracking Index (CTI)	PLC 2		UL 746A
High Amp Arc Ignition (HAI) [6]	PLC 1		UL 746A
Hot-wire Ignition (HWI)	PLC 3		UL 746A

Flammability	Nominal Value	Unit	Test Method
Flame Rating (1.2 mm)	HB		UL 94

Injection	Nominal Value	Unit
Drying Temperature	100 to 110	°C
Drying Time	3.0 to 4.0	hr
Suggested Max Moisture	0.020	%
Suggested Shot Size	30 to 80	%
Hopper Temperature	60 to 80	°C
Rear Temperature	250 to 290	°C
Middle Temperature	255 to 295	°C
Front Temperature	260 to 300	°C
Nozzle Temperature	275 to 300	°C
Processing (Melt) Temp	275 to 300	°C
Mold Temperature	60 to 90	°C
Back Pressure	0.300 to 0.700	MPa
Screw Speed	40 to 70	rpm
Vent Depth	0.038 to 0.076	mm

Injection Notes

- Drying Time (Cumulative): 8 hr

Notes

[1] A UL Yellow Card contains UL-verified flammability and electrical characteristics. UL Prospector continually works to link Yellow Cards to individual plastic materials in Prospector, however this list may not include all of the appropriate links. It is important that you verify the association between these Yellow Cards and the plastic material found in Prospector. For a complete listing of Yellow Cards, visit the UL Yellow Card Search.

[2] Typical properties: these are not to be construed as specifications.

[3] 50 mm/min

[4] Type I, 50 mm/min

[5] 1.3 mm/min

[6] Surface

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Where to Buy

Supplier

SABIC
Web: http://www.sabic.com/

Distributor

3Polymer (Guangzhou) Chemical Technology Co., Ltd.
Telephone: +86-20-3466-7988
Web: http://3polymer.com
Availability: China

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